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Research paper

Carry save adder and carry look ahead adder using inverter chain based coplanar QCA full adder for low energy dissipation

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ABSTRACT

Quantum-dot cellular automata (QCA) technology offers the potential option for performing transistor-less calculations at the nanoscale. In this paper, we introduce a new full adder structure using an inverter chain on a single layer. By utilizing the proposed full adder structure, we design and implement a carry save adder and carry look-ahead adder. Some of these techniques like pipeline structures or asynchronous timing becoming more attractive and are gaining more attention than other solutions. This paper mainly focuses on the high circuit density and energy-efficient aspects of QCA circuits. The design exploits the inherent pipeline nature of QCA, which can lead to an enormous reduction in area using an inverter chain since all computations can be computed in a single block. A comparison among our results and typical structures shows that the presented structures outperformed the best existing designs with respect to area, latency, and cell count. Our structures are more suitable elements for realizing complex QCA circuits with very high operating speed. Functional verification and energy consumption analyses were conducted by well known tools.

1. Introduction

Low power circuits are now tremendously significant in upcoming integrated circuits. One of the most striking features of this emerging technology is that it has the ability to dynamically reconfigure or redesign the functionality of a system, which makes the system more powerful and efficient in terms of computational speed and power dissipation [1]. The focus of this paper is on emerging nanotechnologies, which would fall under the category of new devices. Emerging nanotechnologies offer an intriguing alternative as they inherently operate at low power. The majority of these emerging nanotechnologies tend to be based on digital logic.

In today's computers, binary information is encoded using current switches. The on and off states of current switches are described as “1” and “0” bits, respectively [2]. One of the most promising nanotechnologies is quantum-dot cellular automata (QCA), which was initially proposed by P.D. Tougaw and C.S. Lent in the early 1990's [3].

Since then, a significant quantity of research has focused on QCA, both theoretically and experimentally. This has become one promising candidate for use in nanocomputing. The fundamental component of circuit execution is a QCA cell that is extremely compact, and therefore facilitates extreme densities. Each technical and exploratory investigation on QCA determines which QCA circuits can perform at high operating frequency (*THz*) wavelengths using minimal energy expenditure [4].

We introduce an efficient technique, which is used to overcome the traditional problems of power consumption in any adder based design. Some of these techniques like pipeline structures or asynchronous timing becoming more attractive and are gaining more attention than other solutions. A full adder (FA), carry save adder (CSA) and carry look ahead adder (CLA) were used in this study. The design exploits the inherent pipeline nature of QCA, which can lead to an enormous reduction in area using an inverter chain since all computations can be computed in a single block. One can design highly energy efficient circuits, as QCA does not involve the physical movement of charge. This design will be highly energy efficient, along with the added advantages of pipelining and the small footprint of these devices.

The rest of the paper is organized as followed. The fundamentals of QCA in nanoprocessing are introduced in Section 2. The concepts of the proposed FA, CSA and CLA are discussed in Section 3. The proposed layouts and the energy dispersion products considered during the design research are presented in Section 4. The paper is concluded in Section 5.

2. Materials and methods

QCA nanotechnology is still being researched experimentally and has not been commercialized. There are experimental demonstrations of this technology by researchers in metal island, molecular, semiconductor, and magnetic QCA fabrication technologies. The first experimental demonstration of QCA technology used metal island technology. Four quantum dots were fabricated using interconnected aluminum islands. The major limitation of this design is that the metal islands need to be kept at cryogenic temperatures during electron switching as the size of quantum dot is in microns. Semiconductor QCA implementations have been realized in advanced CMOS processes using GaAs/AlGaAs [9].

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Introduced by Craig Lent in 1993, QCA is one of the emerging technologies likely to supersede conventional CMOS technology [3]. In order to differentiate this proposal and the cellular automata models which are used for performing quantum computation, many authors refers to this subject as QCA. They proposed this new technique to be an alternative method for fabricating electronic devices. The concept was very theoretical in the beginning. Many mathematical models were used to prove the feasibility and strengths of QCA technology. QCA facilitates a computational method, which is different from that used in traditional transistor based circuits. QCA makes use of combined quantum dot arrays to manipulate Boolean logic.

The advantages of this technology are as follows:

- High Operating Speed –*THz* range
- Low Power Consumption – 100 W/cm^2
- High Device Density – 1012 devices/cm^2

This section describes the basic QCA operating principles, starting from a single QCA cell, and on to a QCA wire, and culminating with logic gates. The basic clocking scheme is also described.

A high-level diagram of a four dot QCA cell is shown in Fig. 1. Each QCA cell usually covers four electron wells and two electrons. The dots are combined using quantum tunneling boundaries [5]. The Coulombic repulsive force causes electrons to stay in the farthest dots in a cell that matches the minimum power output. Information encoded in QCA cells is transferred due to Coulombic interactions between neighboring QCA cells; the state of one cell influences the state of another [6].

A QCA wire can be viewed as a horizontal series of cells that transmits information from one cell to another. A binary wire is usually separated through different clock pluses to ensure that the signal does not deteriorate, although signals often decay when using a lengthy chain of cells with identical clocking phases. The QCA cells are usually applied near each other to create a sequence of QCA cells. This sequence is known as a QCA wire [5]. Fig. 2 shows a QCA wire transporting a bit "0". Each cell is polarized to "-1" due to Coulombic interaction. One can see that the occupied dots in the first cell distance themselves from the other occupied dots in the cell. The second cell reproduces the logical value from the first cell and passes this on to the next cell. This process creates a domino effect, in which all cells adopt the same polarization. Fig. 2 shows the same wire carrying a bit "1" as well. In other words, putting a value of "0" or "1" in the first input cell propagates the value to neighboring cells as it pushes away nearby electrons.

QCA circuits use clock system to synchronize and manage data transfer. The electron in the electron well requires high potential energy to tunnel through the junction. Thus, clocking plays a very important role in the synchronization and flow of information along a particular direction. QCA circuits typically use a clock with four clocking phases. It can essentially be viewed as a pump that is constantly pumping out data sequentially. The main advantage of a QCA clock system is that it provides power to run the circuit because the quantum cells do not require an external power source. A QCA clock system uses four clock phases to regulate cells. As mentioned before, QCA clocking can be utilized in the wire crossing. A QCA circuit is divided into four clock zones. The Zone-0 cannot meet with Zone-2; similarly, Zone-1 cannot meet with Zone-3. Under this concept, wire crossing can be easily realized in a QCA circuit [7,8]. A QCA clock with four clock pulses is presented in Fig. 3.

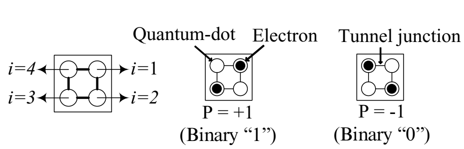

Fig. 1. Schematic of a QCA cell.

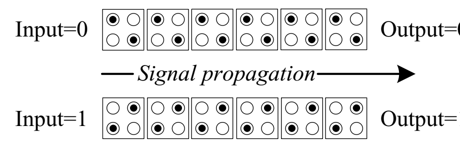

Fig. 2. A binary wire, which propagates information.

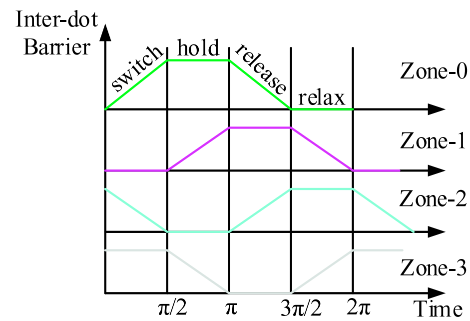

Fig. 3. QCA clock scheme.

2.1. Related research

In this work, we intend to present a CSA and CLA design based on the proposed FA. Before we present details of the proposed adder and the metrics to be used for comparison, we will derive insight into existing adder designs based on QCA.

2.1.1. Background on full adder

The most fundamental mathematical operation is binary addition. The FA and half adder (HA) are widely used for constructing arithmetic circuits. A 1-bit FA has three inputs (A, B, and C) and two outputs (S (Sum) and C(Carry)). C. S. Lent et al. proposed the first QCA FA design in 1994, which consisted of five majority gates and three inverters [7]. It was also designed without a clocking term. The first formulation of the FA in QCA is defined in eqs. (1) and (2):

$$C = M(A, B, C_{in}) \tag{1}$$

$$S = M(M(A, \overline{B}, C_{in}), M(A, B, \overline{C_{in}}), M(\overline{A}, B, C_{in})) \tag{2}$$

where M and M_5 represent 3 and 5-input majority gate respectively. The carry result is obtained from the majority gate. Optimization is required for the Sum equation, because majority gates and inverters generate it. After proposing a new 5-input majority gates, several QCA FAs were proposed [13,17]. The equation for the Sum result is shown in eq. (3). As previously mentioned, the 3-input majority gate plays a fundamental role in nearly all adder designs.

$$S = M_5 - M(A, B, C_{in}, \overline{C_{out}}, \overline{C_{in}}) \tag{3}$$

An FA Boolean logic function considers three inputs (A_i, B_i, C_{in}) and offers two outputs (S_i and C_i) Eqs. (4) and (5) are the sum and carry outputs as a function of the input values. The inputs A_i, B_i, and C_i and the outputs S_i and C_{i+1} are Boolean factors. It can be presumed that A_i and B_i are the i_{th} bits of integers A and B, respectively, and C_i is the carry bit received from the i_{th} position. The FA cell computes the sum bit S_i and carry bit C_{i+1} that will be accepted by the following cell [9].

$$S_i = A_i \oplus B_i \oplus C_i; \tag{4}$$

$$C_{i+1} = A_i B_i + B_i C_i + A_i C_i; \tag{5}$$

2.1.2. Background on carry save adder

The carry save adder (CSA) reduces the addition of three numbers to the addition of two numbers. The total propagation delay is the sum of the propagation delays from three gates, regardless of the number of bits. The carry save unit consists of n full adders, each of which computes a single sum and carries a bit based solely on the corresponding bits of the three input numbers. The main disadvantages of CSA is that sign detection is difficult: When a number is represented as a carry save pair (A, B) such that its actual value is $C + S$, we may not know the exact sign of total sum $C + S$. Unless the addition is performed in full length, the correct sign may never be determined. In that case, CLA is more useful for overcoming this issue. Therefore, comparisons with other type of adders will be presented later.

2.1.3. Background on QCA carry look ahead adder

CLA layout is based on the concept of searching lower adder bits and the addend if greater requests produce a carry bit. This particular adder decreases the carry delay by decreasing the quantity of gates by producing a propagating carry signal. Two Propagate (P) and Generate (G) signals were developed solely for the purpose of decreasing the amount of time involved in producing the carry bits. P and G are propagated through a minimum immense bit placement, regardless of whether a carry bit is produced. Generally, it is possible to determine which P is the sum from the FA that equals zero. However, $A \oplus B$ although create will be the carry out that is A•B.

3. Proposed structures

3.1. Proposed full adder

A new highly area efficient coplanar QCA FA design based on the eqs. (1) and (4) is proposed in this section. To design our improved FA, we used XOR [10] and cell based inverter gates. As shown in Fig. 4(b), the layout of the proposed circuit has 61 cells and requires two clock pulses to generate the *SUM* and *CARRY* outputs. In this circuit, A and B are two 1-bit inputs, and C is the carry input. The proposed QCA, FA uses conventional QCA cells. The use of a compact *XOR* gate [10] results in a circuit that is much simpler than designs that use only 3-input majority gates and inverters. This layout includes extended vertical wires that could result in unreliable signals.

3.2. Proposed carry save adder design in QCA

As mentioned before, the primary advantage of CSA is in the reduced output value. Accordingly, the suggested CSA has twelve input and six output values. This circuit demands four proposed FAs and a ripple carry adder (RCA). That layout involves the lower quantity of QCA cells. Within this section, we will also demonstrate the advances of our proposed circuit. The structure is composed of a regular QCA cell. All input cells are placed on one side, and the outputs are placed on the other side. This helps actualize the proposed design by combining various QCA designs, such as arithmetic and logic unit architecture. A block diagram for the proposed

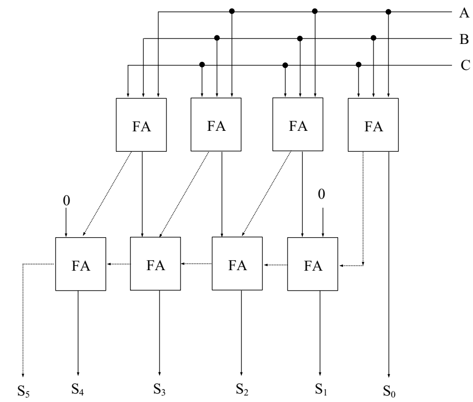

Fig. 5. The block diagram of the proposed CSA.

CSA is shown in Fig. 5. Each bit of the two numbers is passed through a full adder, and the intermediate result is converted to a ripple carry adder to extract the final result. The logics are same as the eq. (1) and (4) however we show a quick example for easier understanding.

A				1	0	1	1
B				1	1	0	1
C	+			1	1	0	1
S				1	0	1	1
C	+		1	1	0	1	
Sum		1	0	0	1	0	1

The proposed CSA uses a low complexity FA and RCA is composed of 696 cells, each with an area of 0.66 μm^2 as shown in Fig. 6. The desired output is generated after 2.25 clock cycles. The new proposed CSA can be substantiated regarding the number of gates, which guarantees lower propagation delay than existing adders. We have demonstrated the layout involving an FA and executed this utilizing just a CSA.

3.3. Proposed carry look ahead adder design using QCA

The CLA is based on computing the carry bits C_i prior to summation. The CLA logic makes use of the relationship between the carry bits C_i and the input bits A_i and Bi. However, the CLA calculates the carry bit in advance to avoid propagation delay. Each bit, the (S_i) is individual regarding latest sum consequences therefore the ripple effect offers subsequently become ended. Nevertheless, caused by growing amount of expense gates during these circuits, propagation delay enhances however even this specific group is efficient as compared to the first one in procedure below particular issue.

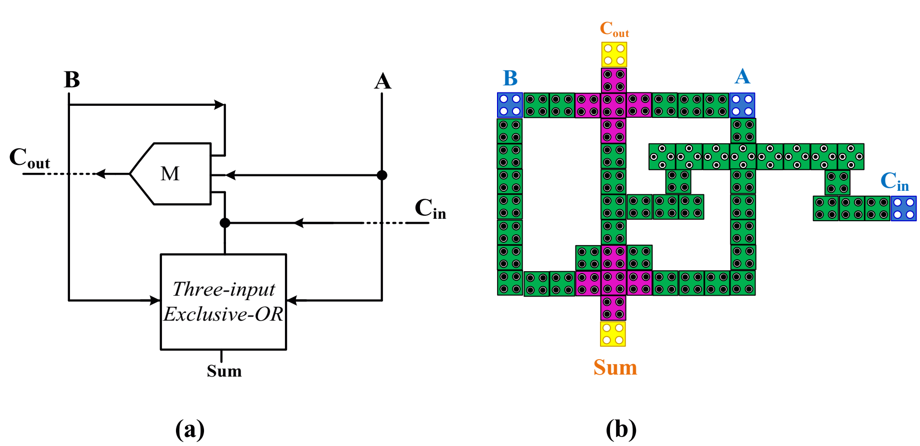

Fig. 4. Proposed FA. (a) Circuit diagram and (b) QCA representation.

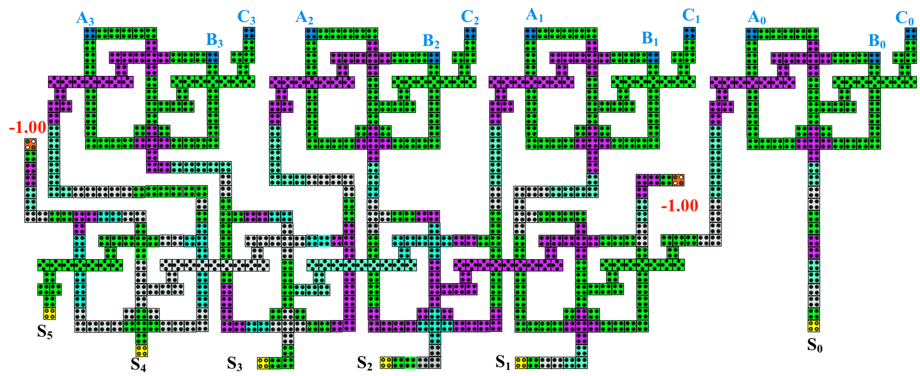

Fig. 6. QCA implementation of the proposed CSA.

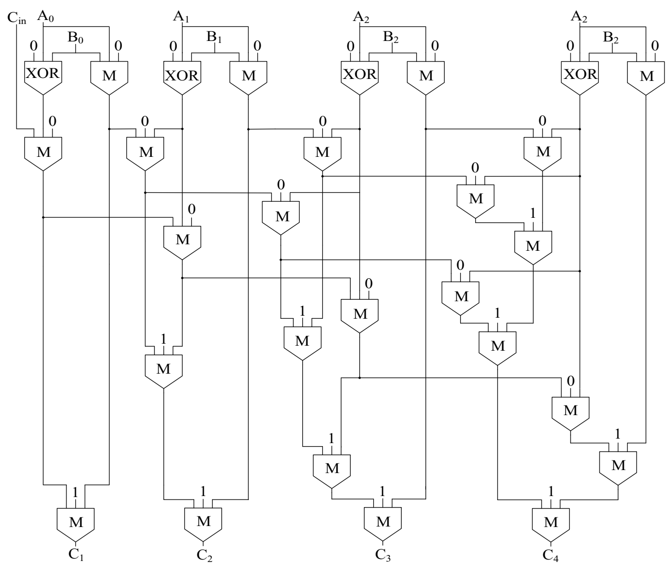

Fig. 7. The block diagram of the proposed CLA.

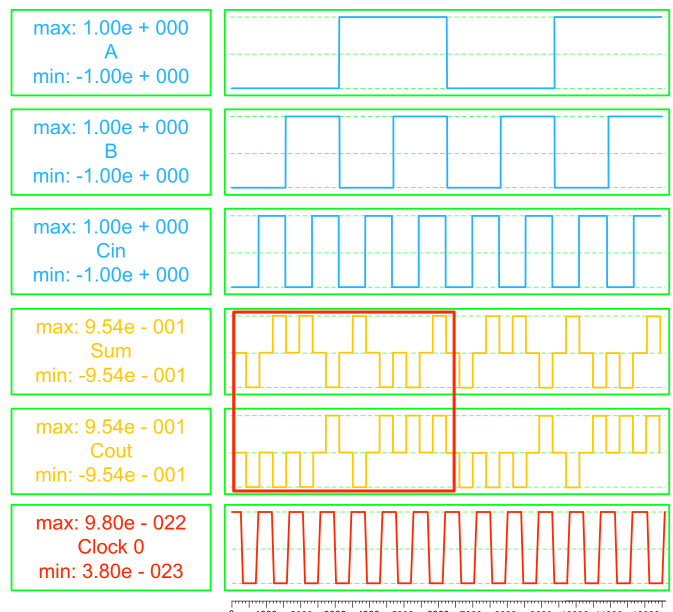

Fig. 9. The simulation results of proposed FA.

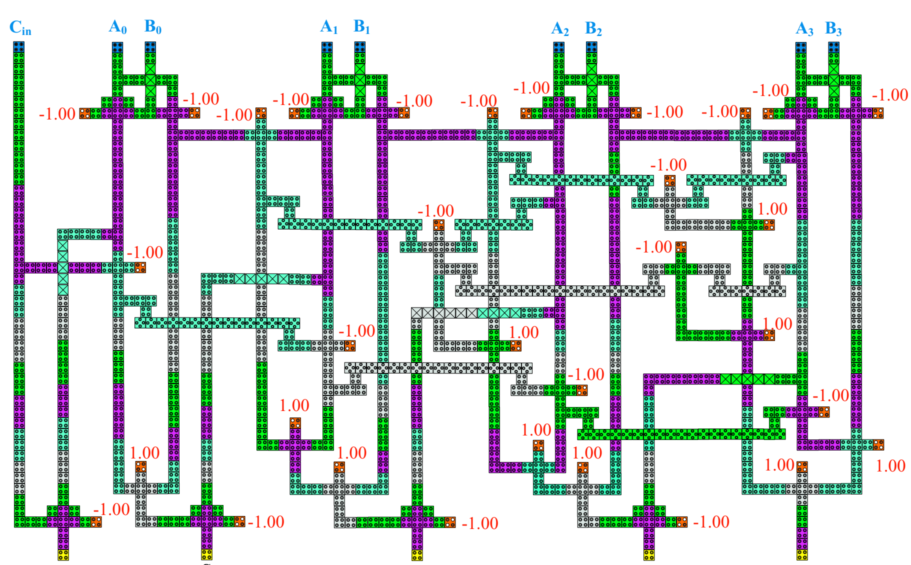

Fig. 8. QCA implementation of the proposed CLA.

Table 1
Complexity comparison of the proposed FA.

Circuit	Complexity (cell count)	Area (μm^2)	Delay (clock cycle)	Cross wiring	Overall Cost
[13]	111	0.13	2.75	Yes	0.9831
[14]	93	0.09	1.25	Yes	0.1406
[15]	79	0.07	1.25	Yes	0.07
[16]	69	0.07	1	Yes	0.07
[17]	63	0.05	1	Yes	0.07
Proposed FA	61	0.04	0.50	No	0.01

$$G_i = A_i \bullet B_i; \tag{6}$$

$$P_i = A_i \oplus B_i \approx A_i + B_i; \tag{7}$$

The equations below define the carry bits in a 4-bit generalized CLA circuit. We can compute all the carry bits in an n-bit adder directly from the auxiliary signals and C_{in} using AND and OR logic operations. The difference between$(A_i \oplus B_i)$and$(A_i + B_i)$ in the truth table occurs when $(A_i = B_i = 1)$. However, if both $A_i and\ B_i$ are equal to binary one, then the G_i term is also one (since its equation is$(A_i \cdot B_i)$), and the$(P_0 \cdot C_0)$term becomes irrelevant as you see the eq. (8). Therefore, OR logic is only an alternative option in the CLA.

$$C_1 = G_0 + P_0\, C_0$$

$$C_2 = G_1 + P_1\, C_1 = G_1 + P_1\, G_0 + P_1 P_0 C_0$$

$$C_3 = G_2 + P_2\, C_2 = G_2 + P_2\, G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$

$$C_4 = G_3 + P_3 C_3 = G_3 + P_3 g G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 \tag{8}$$

The equations essentially indicate that a carry is generated by auxiliary signals and realized by tree networks, as shown eq. (8). Here, $C_0$and C_4 are C_{in} and C_{out} signals in the 4-bit adder, respectively, as illustrated in Fig. 7. The layout of the proposed CLA adder consists of 25 conventional majority gates and five robust QCA inverters, which is shown in Fig. 8. We implemented robust QCA inverters in the structure to obtain a strong signal throughout the wires. The design uses 777 regular QCA cells, and the output is generated after eight clock pulses. The structure occupies an area of approximately 0.92 μm^2. We used a multilayer crossover bridge in the proposed design and in our multilayer wire crossing to obtain a stronger signal and make the structure more stable. Therefore, these can be easily accessed for other purposes required for adder operations. Simulations were performed using the QCADesigner [11], tool as shown in Fig. 11.

4. Results and discussion

We present a detailed power analysis of our proposed approach and compare it with the results obtained from existing designs in this section. We have considered both the upper and lower power limits in order to more accurately capture the advantages offered by our approach. Before we present a comparison between the proposed approach and existing designs, a detailed analysis between the proposed scheme and the existing conventional scheme based on power bound models is presented in

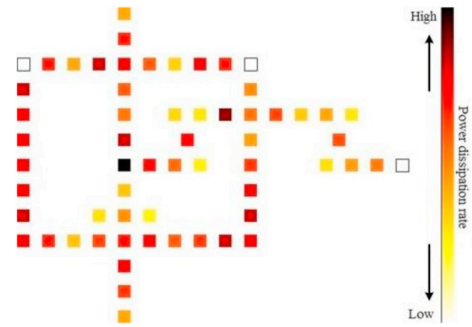

Fig. 10. The power dissipation maps of FA with 0.5 E_k.

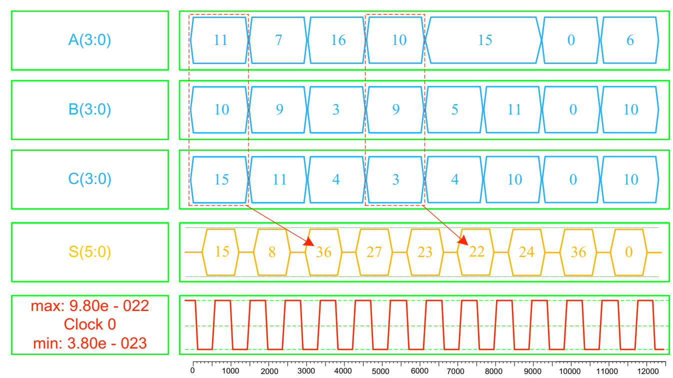

Fig. 11. Simulation result for the proposed CSA.

this section. We attempted to compute the upper and lower bounds for both schemes and compare the power efficiency results.

We evaluate and compare the existing QCA structures for use in an FA, CSA, and CLA in terms of structural and energy consumption in order to select the best design for leveraging our proposed adder structures. The simulation setup was developed for two multi-gate functionality circuits. The simulation was performed with QCADesigner 2.0.3. The proposed circuits were simulated and the outcomes were analyzed in order to determine the effects of different QCA parameters. The proposed FA works properly for circuits with low power consumption. The simulation results are shown in Fig. 9.

The proposed QCA FA structure design was optimized with QCA technology. This circuit was designed to include the lowest possible number of cells with the help of cell minimization methods. The techniques use a rotated cell based inverter. Table 1 shows the advantage of the proposed design compared with the designs in [13–17] in terms of circuit complexity. Moreover, the presented designs are the only designs with simultaneous single layer accessibility to the input and output cells. The design includes a single XOR gate [10], 61 cells with area of 0.04 μm^2 each, and two clock pulses, which is significantly better than earlier designs. We compared the proposed FA structure with previous designs presented in [13–17].

As mentioned earlier, energy dissipation in the QCA circuits can be estimated with the QCAPro tool [12]. To demonstrate the energy consumption in different QCA structures for use in an FA, a detailed energy

Table 2
Energy dissipation comparison between the proposed FA and earlier designs with different tunneling energy levels.

The circuits	Avg. leakage energy dissipation (meV)			Avg. switching energy dissipation (meV)			Avg. energy dissipation of circuit (meV)		
	0.5 E_k	1.0 E_k	1.5 E_k	0.5 E_k	1.0 E_k	1.5 E_k	0.5 E_k	1.0 E_k	1.5 E_k
[13]	0.036	0.096	0.1926	0.1359	0.1126	0.1296	0.1923	0.2037	0.2922
[14]	0.046	0.091	0.1871	0.1590	0.1404	0.1018	0.1855	0.2217	0.2894
[16]	0.034	0.098	0.1717	0.1385	0.1196	0.1043	0.1731	0.2094	0.2660
[17]	0.039	0.098	0.1638	0.1370	0.1231	0.1100	0.1667	0.1997	0.2539
Proposed FA	0.029	0.086	0.1516	0.1279	0.1081	0.0903	0.1575	0.1947	0.2420

Table 3
Complexity comparison of the proposed CSA.

Circuit	Complexity (number of QCA cells)	Area(μm^2)	Delay (clock cycle)	Layer type	Overall Cost
Kogge-Stone [18]	815	0.738	4.00	Multilayer	11.808
Ladner-Fischer [18]	698	0618	4.00	Multilayer	9.888
CSA [19]	525	0.55	2.50	Multilayer	3.44
Proposed CSA	696	0.66	2.25	Coplanar	3.34

Table 4
Complexity comparison of the proposed CLA.

Circuit	Complexity (number of QCA cells)	Area(μm^2)	Delay (clock cycle)	Layer type	Overall cost
CLA [20]	1575	1.89	3.50	Coplanar	23.15
CLA [21]	1758	1.89	4.00	Coplanar	30.24
Proposed CLA	777	0.92	2.75	Multilayer	6.95

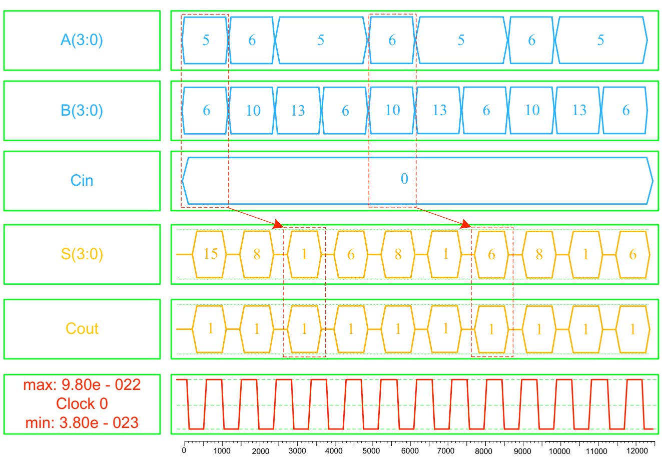

Fig. 12. Simulation results for the proposed CLA.

analysis was conducted for different tunneling energy levels. By considering the average leakage and switching energy dissipations, the proper average energy consumption of the circuits can be achieved. The energy dissipation of the proposed QCA FA circuit was estimated using QCAPro tools at temperature $T = 2.0\,K$ at various tunneling energy levels. The results from a comparative energy dissipation study between the proposed QCA FA and earlier designs [13–17] are listed in Table 2. Power dissipation maps in the proposed QCA FA with 0.5 E_k are shown in Fig. 10.

In the past few sections, we have clearly explained how the proposed 1-bit adder performs n-bit computations without the need for stacking of 1-bit adders, which is the case in traditional adders. This gives us an enormous advantage with regard to area, as the number of QCA cells remains constant. The number of cells is approximately 61, 525 and 777 in our proposed FA, CSA and CLA respectively. Simulation result for the QCA, CSA are shown in Fig. 11. QCA cells have small footprints in digital circuits. They typically have an area of 100 nm^2 with a cell dimension of 10 nm. In this section, performance of the proposed CLA is also discussed. The QCA adder structure was implemented and characterized in the previous section. This work deals with designing CSA and CLA circuit using a QCA. This design is composed of the proposed compact FA to implement the logic structure. A CLA can be designed by integrating the majority gates. The advantage of using this QCA technology is its reduced size and increased execution speed with considerably reduced power consumption.

As one can see in Table 3, the area occupied by the proposed CSA using RCA is less when compared to other designs [18,19]. This is because the proposed CSA uses a compact full adder to increase the operation speed, along with less area. Note that the multilayer circuitry also consumes additional power and layers in fabricated circuits. We have demonstrated advances in another metrics too, including latency. Therefore, the overall cost of the proposed complex adders is decreased significantly compared to previous designs [18,19].

One can see in Table 4 that the proposed CLA shows enhancements above 50% in key metrics compared to the best and recent circuit presented in design of [20] while maintaining low power consumption. Compared to previous adders [20,21], the best adder is our proposed CLA because it requires less space and consumes less power. Simulation result for the QCA, CSA are shown in Fig. 12.The space and power consumption of the proposed circuit is much lower, and the device has lower delay compared to the designs in [20,21]. Since the suggested CLA uses fewer QCA cells, it has a simple layout and requires less wiring in comparison to other adders.

The goal of this work was to exploit the robust compact nature of QCA adders and design energy efficient circuits. A custom area efficient technique based on switching the polarity of the FA was established during the initial phase of this research. The efficiency and gains provided by the custom technique were demonstrated with the help of an XOR gate [10] based on low power consumption when compared with existing designs.

5. Conclusion

The objective of this paper is to take advantage of the energy efficiency of QCA adders and design low power QCA circuits. The main part of the study focused on the implementation of a high performance FA. Power dissipation analysis showed that the proposed circuit would be useful for circuit designers in providing a more complete view of QCA operations. A compact and robust CSA is proposed in this study. The circuit is compared with the best existing designs, and our proposed circuit is more efficient. We also introduced a CLA structure based on the proposed FA. The proposed scheme offers appreciable logic depth with only a few extra components while considering carry propagation. The implementation of the proposed 4-bit CLA scheme has been shown to have smaller area than the implementation of existing CLA designs. It has been verified that the implemented CLA can also operate at higher speeds than the implemented FA, with both adders designed with equal sized transistors. Furthermore, the results have shown that the implemented CLA has lower delay and the number of cell.

Acknowledgments

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